Cognitive Equation Wrapper System

Functional Architecture

Purpose

The Cognitive Equation Wrapper System organizes established computational models into a connected functional architecture.

It does not replace the underlying models or claim that any single equation explains cognition. Instead, it gives each model a functional role within a larger story: how information is detected, represented, remembered, predicted, evaluated, acted upon, and used to update future behavior.

The wrapper is workflow-agnostic by design. It can be introduced as a preprocessing layer, post-processing layer, monitoring layer, audit layer, or decision-support layer without requiring replacement of the host system.

The Functional Story

1. Observation

The environment produces information, but the system never receives a perfect representation of reality.

Signals differ in strength, timing, relevance, clarity, and noise. The first function is therefore to determine what can be detected and what should be treated as meaningful.

Story: The world produces signals. The system receives imperfect observations. Before anything can be believed, remembered, or acted upon, the system must determine what is actually being detected.

2. Representation

Detected information must be converted into an internal representation.

The underlying state of the environment or organism may not be directly observable. The system therefore constructs an estimate of what may be happening behind the available observations.

Story: Observation is not the same as state. The system must infer what is happening behind what it can directly observe.

3. Memory

The current representation is influenced by retained history.

Information does not remain equally available over time. Previous exposure, retrieval, timing, and reinforcement influence which parts of the past remain relevant to the present.

Story: The present is informed by the past, but the past is not perfectly preserved. The system must continually determine what remains available and what has become less accessible.

4. Prediction

Once a representation exists, the system can anticipate what may happen next.

Prediction allows the system to move beyond passive reception and create an expectation against which future observations can be compared.

Story: The system asks what should happen next. It creates an expectation from its current model, then waits to see whether reality agrees.

5. Prediction Error

When reality differs from expectation, a discrepancy appears.

That discrepancy identifies a possible weakness in the current model. A surprising event may therefore become more informative than an unsurprising event because it reveals what the system failed to anticipate.

Story: Expected and observed outcomes diverge. The difference becomes informative and may signal that the current model needs to change.

6. Uncertainty

The system must represent not only what it believes, but how uncertain that belief is.

Several competing explanations create a wider range of possibilities. Converging evidence narrows the range and concentrates belief around fewer explanations.

Story: The system does not only ask, “What do I think?” It also asks, “How uncertain am I?”

7. Belief Updating

New evidence changes the relationship between competing explanations.

A belief is not necessarily fixed. It can be revised as evidence accumulates, especially when new information is more consistent with one explanation than another.

Story: A prior belief encounters new evidence. The evidence changes the relative plausibility of possible explanations, producing an updated belief state.

8. Hypothesis Competition

Evidence becomes more meaningful when it is evaluated against alternatives.

The system does not ask only whether an observation is compatible with one hypothesis. It asks whether the observation supports one explanation more strongly than competing explanations.

Story: Explanation A predicts one pattern. Explanation B predicts another. New evidence changes the competition between them.

9. Information Value

Once uncertainty exists, the system can evaluate information according to its potential usefulness.

Some observations barely change the current belief state. Others sharply distinguish between competing possibilities.

Story: Not every observation is equally useful. Some information merely confirms what is already expected; other information changes the entire interpretation of the situation.

10. Attention

Cognitive resources are limited, so available information cannot receive equal processing priority.

Attention determines which signals, representations, or hypotheses receive greater weight at a given moment.

Story: Many signals arrive at once. The system cannot process all of them equally. Attention determines what receives priority.

11. Resource Allocation

Attention operates within finite processing capacity.

Competing tasks, excessive information, irrelevant signals, and complex representations consume resources that might otherwise be available for the central problem.

Story: Processing capacity is finite. Every additional demand competes for that capacity. The system must allocate limited resources among competing requirements.

12. Evidence Accumulation

Information can accumulate over time rather than producing an immediate decision.

The system may continue gathering evidence until the available support becomes strong enough to justify action or strong enough to justify waiting.

Story: One observation may not be enough. Evidence accumulates sequentially until the system reaches a meaningful level of support.

13. Decision Threshold

A decision requires a criterion for determining when enough evidence has been collected.

A lower threshold may support faster action. A higher threshold may support greater caution but create additional delay.

Story: The system must decide when to stop investigating. Too little evidence can produce premature action; too much required evidence can produce unnecessary delay.

14. Valuation

Evidence alone does not determine action.

Possible outcomes have different consequences. The system evaluates those consequences according to expected benefit, cost, risk, urgency, or subjective importance.

Story: The system identifies possible outcomes and evaluates what each outcome would mean. Action depends on both what appears likely and what matters.

15. Action Selection

Multiple possible actions compete according to their estimated value, probability, uncertainty, and consequences.

The selected action may be deterministic or probabilistic, depending on the system and the surrounding conditions.

Story: Several actions are available. Each receives an estimated value. The system selects among them according to the relative strength of the alternatives.

16. Exploration

The system may sometimes choose an option that is not currently believed to be optimal because it could produce valuable new information.

This creates a tension between using what is already known and investigating what remains uncertain.

Story: The best-known option is not always the best next experiment. An uncertain option may teach the system something important.

17. Outcome

An action produces consequences that affect the subsequent state.

The significance of an action extends beyond its immediate result because the outcome changes what becomes possible, valuable, or likely next.

Story: A decision becomes an event in the environment. The environment responds. That response becomes the next piece of evidence.

18. Learning

The outcome is compared with what was expected.

The difference can modify future predictions, beliefs, values, priorities, and action choices. Learning connects past decisions to future behavior.

Story: The system acted. Reality provided an outcome. The difference between expectation and outcome changes what the system will expect or choose next time.

The Wrapper Structure

The architecture is not a rigid eighteen-step pipeline. It is a routing and relationship layer that allows functions to interact within recurring cycles.

The primary cycle moves from observation to representation, memory, prediction, comparison, belief updating, decision, action, outcome, and learning. The updated state then influences the next observation.

The uncertainty cycle moves from uncertainty to information value, attention, evidence collection, and reduced uncertainty.

The learning cycle moves from action to outcome, prediction error, learning, and future action.

These loops may operate at different speeds and levels. A system may update a belief immediately, accumulate evidence over time, revisit memory later, or change its decision threshold depending on context.

What the Wrapper Adds

The equation library contains established mathematical references. The wrapper gives those references a new functional story.

For each reference, the wrapper asks:

- What function does this model support?
- What does the function receive?
- What does it produce?
- Which other function can use that output?
- What loop does it participate in?
- How could the relationship be evaluated?
- Where can the function be inserted into an existing workflow?

The core transformation is:

> Established models become functional components.
> Functional components become relationships.
> Relationships become cognitive loops.
> Cognitive loops become an adaptive system.

That is the proposed contribution. The novelty is not the invention of every underlying equation. The novelty is the organization, routing, interpretation, and creation of new functional stories from established computational references.

Academic Positioning

The system should therefore be presented as:

> “A proposed functional and relational wrapper for organizing established computational models into recursive cognitive workflows.”

Ah — **riddle the architecture instead of explaining it outright.** That actually fits what you're building much better.

The Cognitive Wrapper — Riddled

I receive what I cannot fully see.
I remember what I cannot perfectly keep.
I predict what has not yet happened.

When reality disagrees with me,
I do not necessarily fail.
I have received information.

I ask:

What do I believe?
How certain am I?
What would change my mind?
Where should I look next?

I cannot examine everything.
I cannot remember everything.
I cannot calculate every possibility.

So I choose.

I gather evidence until something crosses a boundary.
I compare possibilities.
I assign value to consequences.
I act.

Then the world answers.

Sometimes it confirms me.
Sometimes it surprises me.
Sometimes it teaches me that the question itself was wrong.

And so I change.

The next time I see the world,
I am no longer exactly the system that saw it before.

What am I?

A system that turns observation into belief, belief into prediction, prediction into error, error into learning, and learning back into observation.

But there is another question:

If every mechanism inside me can be replaced,
yet the relationships between them remain—
what is the thing that remains?

That is the wrapper.

And the really fun part is the hidden structure:

Observe → Represent → Remember → Predict → Surprise → Update → Question → Attend → Decide → Act → Learn → Repeat.

The equations are the **clues**.
The cognitive functions are the **answers**.
The relationships are the **riddle**.

DB3: Synthetic Reasoning and Analysis Layers
Cognitive Equation Wrapper Extension

DB3 transforms evidence in DB2 into structured analytical outputs such as summaries, comparisons, pattern recognition, ranked interpretations, and hypothesis generation. It is a read-only synthesis layer rather than a source of truth: its outputs are derived from the available evidence and may change when the evidence base changes.

Within this extension, DB3 provides a model-agnostic wrapper for established mathematical and computational mechanisms associated with observation, representation, inference, uncertainty, attention, memory, prediction, decision-making, learning, and adaptation. Individual equations or models are treated as analytical components rather than as claims that human cognition literally implements any single mathematical formulation.

The wrapper does not replace the governed deterministic core. It provides a structured interpretation layer through which heterogeneous analytical mechanisms can be composed, compared, evaluated, and traced while preserving the underlying requirements for provenance, governance, validation, and auditability.

DB3 should include:

- Pattern-detection outputs.
- Comparative analyses.
- Structured interpretations.
- Ranked hypotheses or candidate explanations.
- Confidence and uncertainty indicators.
- Prediction and prediction-error observations.
- Information-value assessments.
- Attention or prioritization signals where applicable.
- Decision-support outputs.

• Learning or state-update observations where applicable.
• Traceable links back to DB2 evidence.
• Model and method provenance.
• Versioned analytical outputs.
• Explicit separation between source evidence and derived interpretation.

The cognitive wrapper is organized around a recurring transformation:

Observation
→ Representation
→ Prediction
→ Comparison
→ Update
→ Information Selection
→ Decision
→ Action
→ Outcome
→ Learning
→ Updated State

This sequence is not required to operate as a single linear pipeline. Individual components may form feedback relationships, operate conditionally, or be omitted depending on the analytical task.

The underlying mathematical mechanisms remain modular. For example:

• Signal Detection Theory — observation and discrimination.
• State-Space Models — latent-state representation.
• Hidden Markov Models — inference over hidden states.
• Bayesian Inference — belief updating.
• Shannon Entropy — uncertainty representation.
• Information Gain — information-value assessment.
• Mutual Information — dependency and information relationships.
• Attention Models — weighting and allocation.
• Memory Models — temporal retention and accessibility.
• Predictive Coding — prediction and discrepancy.
• Prediction Error — model/expectation adjustment.
• Drift-Diffusion Models — evidence accumulation.
• Sequential Testing — evidence sufficiency and stopping.
• Utility Models — consequence evaluation.
• Softmax Choice — probabilistic action selection.
• Reinforcement Learning — adaptive value updating.
• Bellman Models — future consequence evaluation.
• Exploration–Exploitation Models — information versus immediate value.

These mechanisms are not required to be used simultaneously. The wrapper determines which mechanism is appropriate to the analytical function being performed and maintains the relationship between the selected mechanism, its inputs, its outputs, and its originating evidence.

The system's operational lifecycle is mapped across seven non-speculative, empirical loops that provide the universal wrapper. The cognitive mechanisms above are instantiated within these loops rather than creating a separate architecture.

Loop 1 — Input → Interface → Coupling → Transformation → Output

This loop establishes the basic analytical pipeline. Evidence enters through defined interfaces, is coupled to the appropriate analytical representation, transformed by a selected method, and produces a structured output.

Within the cognitive wrapper, this can correspond to:

Evidence
→ Observation
→ Representation
→ Analysis
→ Output

The loop establishes where a cognitive or mathematical mechanism enters the governed system and where its derived result exits.

Loop 2 — Disorder → Pattern → Stability

This loop represents the transition from heterogeneous or variable information toward detectable structure and stable representations.

Within the cognitive wrapper:

Variation / Noise
→ Signal
→ Pattern
→ Representation
→ Stability

Entropy, information-theoretic measures, signal detection, and pattern-recognition mechanisms may provide analytical components within this loop.

The wrapper does not assume that every apparent pattern is meaningful. Pattern outputs remain derived results and must remain distinguishable from source evidence.

Loop 3 — Detection → Diagnosis → Correction → Prevention

This loop represents identification of deviations, interpretation of their likely causes, corrective response, and subsequent refinement.

Within the cognitive wrapper:

Detection
→ Hypothesis
→ Evidence Comparison
→ Error Identification
→ Update

Bayesian inference, hypothesis competition, prediction error, sequential evidence accumulation, and related mechanisms can instantiate portions of this loop.

The important architectural property is traceability: the interpretation must remain linked to the observations that produced it.

Loop 4 — Entropy → Conditioning → Extraction → Differentiation

This loop represents the transformation of uncertain or heterogeneous information into distinguishable analytical structure.

Within the cognitive wrapper:

Uncertainty
→ Conditioning
→ Information Extraction
→ Differentiation
→ Reduced Ambiguity

Entropy, information gain, mutual information, signal-to-noise relationships, and related analytical mechanisms can operate within this loop.

The objective is not simply to increase information quantity. The objective is to determine whether the available information improves discrimination among relevant states, hypotheses, or interpretations.

Loop 5 — Excitation → Coupling → Response → Feedback

This loop represents interaction between an input, a system, its response, and subsequent feedback.

Within the cognitive wrapper:

Input
→ Internal Response
→ Action
→ Outcome
→ Feedback

Prediction, prediction error, state estimation, reinforcement learning, and control-oriented mechanisms can occupy different portions of this loop.

The loop creates the transition from passive analysis toward adaptive behavior while retaining the distinction between analytical output and governed action.

Loop 6 — Data → Model → Prediction → Adjustment

This loop provides the strongest direct connection to the cognitive equation wrapper.

Data
→ Model
→ Prediction
→ Observation
→ Prediction Error
→ Adjustment

The model generates an expectation from available evidence. New evidence is compared against that expectation, and the resulting discrepancy can inform model or state adjustment.

This loop can accommodate Bayesian updating, state-space estimation, predictive coding, reinforcement learning, and related approaches without requiring any single model to be treated as universally correct.

The essential relationship is:

Prediction
→ Difference
→ Update

Loop 7 — Governance → Decision → Validation → Ledger

This loop surrounds the analytical system with governance.

Governance
→ Authorized Analysis
→ Decision / Interpretation
→ Validation
→ Recorded Result

Every derived output remains subject to the governing architecture. Analytical models do not become sources of truth merely because they generate confident outputs.

The wrapper therefore preserves:

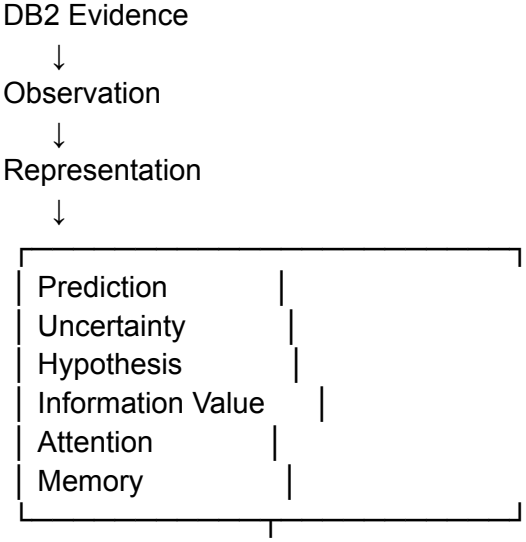
• Provenance.
• Versioning.
• Traceability.
• Validation.
• Human or authorized review where required.
• Policy controls.
• Audit records.
• Separation of evidence from interpretation.
• Reproducibility where applicable.

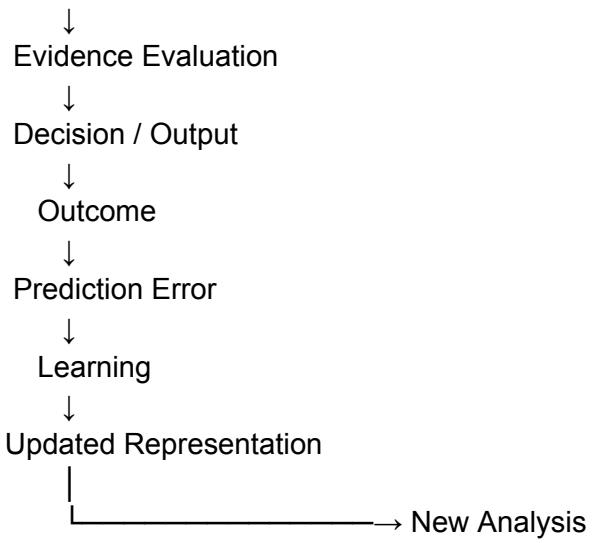
COGNITIVE WRAPPER RELATIONSHIP MODEL

The seven loops provide the stable architecture.

The cognitive mechanisms provide interchangeable analytical components.

Their relationship can be represented as:





This creates a recursive analytical structure without requiring the database itself to become an autonomous cognitive system.

RELATIONSHIP PRINCIPLE

The wrapper treats relationships between mathematical mechanisms as first-class analytical objects.

For example:

Equation / Model A
→ supplies evidence or state
→ to Model B
→ whose output changes the input to Model C.

Thus:

A → B

does not mean that A and B are mathematically identical or causally equivalent. It means that A's output can function as an input, constraint, weighting factor, hypothesis state, uncertainty estimate, or other defined interface for B.

A more general relationship is:

Model A
→ Output
→ Interface
→ Model B

→ Derived Output
→ Validation

This permits relationships such as:

Prediction
→ Prediction Error
→ Learning

Uncertainty
→ Information Value
→ Attention

Attention
→ Selected Evidence
→ Belief Update

Belief
→ Prediction
→ Observation

Action
→ Outcome
→ Prediction Error

Evidence
→ Hypothesis Competition
→ Decision

The wrapper therefore provides a common interface through which different mathematical descriptions can be connected without requiring them to be collapsed into a single theory.

SYNTHETIC REASONING LAYER

The resulting DB3 function is:

Evidence
→ Structure
→ Relationships
→ Interpretation
→ Hypothesis
→ Decision Support

The word “synthetic” refers to the combination of multiple evidence-derived analytical outputs into a structured representation. It does not imply autonomous consciousness, human-equivalent reasoning, or an independently verified model of the human mind.

A DB3 output may therefore state:

Observed evidence
→ detected pattern
→ candidate interpretation
→ uncertainty
→ supporting evidence
→ contradictory evidence
→ predicted consequence
→ recommended analytical next step

Each stage remains distinguishable.

This preserves the fundamental distinction:

SOURCE OF TRUTH:
DB2

INTERPRETATION:
DB3

GOVERNANCE:
CONTROL / POLICY LAYER

RECORD:
AUDIT / LEDGER

DB3 therefore does not overwrite the evidence from which it was derived.

WRAPPER INVARIANTS

Regardless of which analytical mechanisms are inserted, the wrapper preserves several fixed properties:

1. Evidence remains distinguishable from interpretation.

2. Derived outputs remain traceable to their inputs.

3. Analytical models remain replaceable.

4. Model outputs remain subject to uncertainty.

5. The analytical layer does not become the source of truth.

6. Governance remains external to model confidence.

7. Outputs remain versionable and auditable.

8. New modules can be introduced without requiring the underlying deterministic architecture to be redesigned.

9. Different mathematical mechanisms can address the same functional layer and therefore be compared experimentally.

10. A composite result can be decomposed back into the analytical components that produced it.

SUMMARY

DB3 is the synthetic reasoning and analysis layer that transforms governed evidence from DB2 into structured, traceable analytical outputs. The cognitive equation wrapper extends this role by providing a model-agnostic interface for established mathematical mechanisms covering observation, representation, uncertainty, inference, information selection, attention, prediction, decision-making, learning, and adaptation.

The seven universal loops remain the stable architecture:

Input → Interface → Coupling → Transformation → Output

Disorder → Pattern → Stability

Detection → Diagnosis → Correction → Prevention

Entropy → Conditioning → Extraction → Differentiation

Excitation → Coupling → Response → Feedback

Data → Model → Prediction → Adjustment

Governance → Decision → Validation → Ledger

The individual mathematical models are components within those loops, not replacements for them. Their outputs can be connected through defined interfaces, compared, validated, substituted, or removed while preserving the underlying governance structure.

The resulting architecture is therefore a forward-compatible analytical wrapper: a governed space where multiple mathematical descriptions can be composed into structured reasoning while preserving provenance, uncertainty, auditability, and separation between evidence and interpretation.

The central principle is:

Evidence is preserved.
Interpretation is derived.
Models are replaceable.
Relationships are explicit.
Governance remains fixed.